

FCRel at PowerTech Labs

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August 4, 2025

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Dear Dr. Shen,

The enclosed report, titled “*Assembly and Testing of a Small-Scale PEM Fuel Cell Using the Greenlight G20 Test Station*”, was prepared as a requirement for my first co-op work term (MSE 293 – Co-op Work Term 1) in the Summer 2025 semester. The purpose of this report is to document the procedures, challenges, and outcomes of assembling and validating a small-scale PEM fuel cell at FCRel PowerTech Labs.

During my time at FCRel PowerTech Labs in Surrey, British Columbia, I worked as a Research Assistant under the supervision of Dr. Erigene Bakangura, Head of Lab Operations at the Fuel Cell Research Lab. My responsibilities involved both independent and collaborative work related to sensor configuration, leak testing, data acquisition, and fuel cell validation using Greenlight Innovation’s G20 testing system. This report details the full cycle of preparation, testing, and operational procedures carried out during my placement.

I hereby confirm that I have received no further help in writing this report other than what is stated above and that this report has not been previously submitted for academic credit at Simon Fraser University or any other institution.

Sincerely,



Thanh Nam Tran

Student ID: 301540228

Assembly and Testing of a Small-Scale PEM Fuel Cell Using the Greenlight G20 Test Station

Co-op Technical Report – Work Term 1

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Abstract

This report presents the assembly and initial testing of a small-scale proton exchange membrane (PEM) fuel cell at FCReL PowerTech Labs, utilizing the Greenlight Innovation G20 test station. The objective was to conduct a full operational cycle—beginning with the mechanical and electrical assembly, through gas system integration and leak testing, to test execution using the Emerald FCATS control software.

As part of Fuel Cell Research Lab, the project involved configuring a single-cell fixture using a membrane electrode assembly (MEA), gas diffusion layers (GDLs), and custom 3D-printed hardware. Thermocouples, heater wires, and ring terminals were integrated into the fuel cell setup, followed by installation on the G20 platform. System safety was ensured via nitrogen and hydrogen leak detection using both liquid and electronic methods.

Test automation was achieved through the FCATS scripting interface, allowing precise control of flow rates, temperature, humidity, and cell current. Key parameters such as voltage, current density, temperature distribution, and relative humidity were continuously monitored and logged.

The assembled fuel cell operated as expected under the startup and conditioning protocols. No leaks were observed after system pressurization, and all data acquisition tools responded correctly. These results validate the integrity of the assembly procedure and demonstrate the capability of the G20 platform for controlled PEM fuel cell testing.

This work contributes to FCReL PowerTech's fuel cell testing infrastructure and provides a reproducible framework for future conditioning and performance experiments involving advanced MEAs. It is recommended that future efforts focus on automation of sensor calibration and improved thermal insulation of the wiring assembly. Representative performance data is included to illustrate typical cell behavior; full data sets are archived internally for reference.

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Background

PEM Fuel Cells

A proton exchange membrane (PEM) fuel cell converts hydrogen and oxygen into electricity, heat, and water through an electrochemical reaction. Hydrogen at the anode splits into protons and electrons; protons cross the membrane, electrons flow through an external circuit, and at the cathode they recombine with oxygen to form water. PEM cells operate at low temperature, offer high power density, and start quickly—suited to transportation, stationary, and portable power.

Why Test Fuel Cells

Controlled testing quantifies voltage–current behavior, efficiency, thermal response, and degradation. It validates designs, reveals failure modes, and sets operating limits to guide product integration.

G20 Test Station

The Greenlight Innovation G20 provides closed-loop control of gas flow, humidity, temperature, back-pressure, and electrical load, with integrated sensing, real-time logging, and automated sequences via Emerald FCATS. Safety features include interlocks, emergency stops, automated purges, and software limits. Figures 4–5 show the installed system and facility connections.

Glossary of Common Terms

- **PEM** – Proton Exchange Membrane
- **DUT** – Device Under Test (the fuel cell)
- **Anode/Cathode** – Electrodes where reactions occur
- **FCATS** – Fuel Cell Automated Test System
- **MEA** – Membrane Electrode Assembly
- **OCV** – Open Circuit Voltage (no load applied)

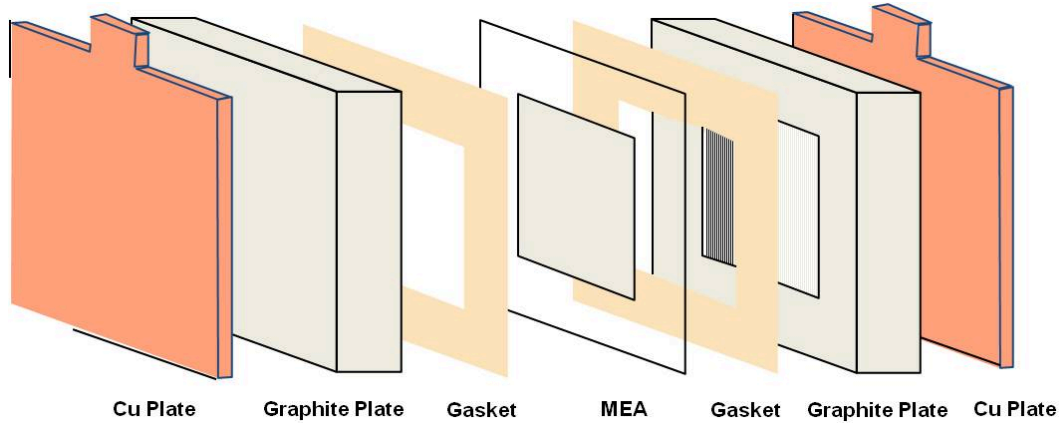


Figure 1: Exploded view of a typical PEM fuel cell assembly.

1. Introduction

This report presents the assembly and validation of a single-cell proton exchange membrane (PEM) fuel cell using the Greenlight Innovation G20 test station at FCREL PowerTech Labs. The project aimed to develop a concise, repeatable procedure for safe operation, controlled testing, and reliable data collection to support future MEA studies.

1.1 Problem Statement

Inconsistent assembly steps, incomplete leak checks, and unverified sensor calibration can lead to unsafe conditions and unreliable data. FCREL required a standardized procedure integrating mechanical assembly, gas handling, wiring, and software configuration.

1.2 Purpose and Objectives

The purpose of this work was to implement and verify such a procedure. The objectives were to:

1. Assemble the single-cell fixture and instrumentation.
2. Mount the assembly to the G20 and connect gas/electrical interfaces.
3. Perform nitrogen and hydrogen leak checks.
4. Configure Emerald FCATS and run startup/conditioning scripts.
5. Log key operating parameters and confirm stable cell performance.

2. Methodology

This section outlines the step-by-step procedures used to assemble and test a single-cell PEM fuel cell using the Greenlight Innovation G20 test station. All work was performed in accordance with lab safety standards and the Standard Operating Procedure (SOP) developed internally at FCREL PowerTech Labs.

2.1 Tools, Equipment, and Materials

The following tools and materials were prepared prior to assembly:

- Torque screwdriver
- Isopropanol (IPA) for cleaning
- Kapton tape for insulation
- 3D-printed sensor fixtures (white plastic holders)
- Hose clamps and mounting hardware
- Fuel cell wiring harness with thermocouples and ring terminals
- Membrane Electrode Assembly (MEA)
- Gas Diffusion Layers (GDL – carbon paper)
- Hydrogen gas: Supplied from a 16-cylinder pack of 99.999% ultra-high purity (UHP) H₂, connected via fixed manifold to the lab's internal line.
- Nitrogen gas: Supplied from a 230 L liquid nitrogen Dewar (Linde) with gas-phase vaporizer output.
- Both gases regulated to ~10–20 psi delivery pressure using dual-stage regulators before entering the fuel cell system.
- Thermal insulation foam
- Leak detection fluid (CANTESCO Formula 300)

- Bacharach Leakator Jr. (gas sniffer)
- Greenlight G20 Test Station
- Emerald FCATS control software
- Gamry Reference 3000 Potentiostat

2.2 Sensor and Fixture Assembly

The wiring harness and thermocouples were integrated into the 3D-printed fixture:

- Sensor wire ends were stripped and inserted into black thermocouple terminal blocks.
- Wire junctions were insulated using Kapton tape to prevent shorts and reduce interference.
- The entire harness was mounted into the fixture using hose clamps to ensure mechanical stability.
- Connections were color coded:
 - Red wires → Cathode
 - Black wires → Anode
- Gas lines were also color-coded:
 - Red (uninsulated) → Cathode
 - Black (insulated) → Anode
- MEA orientation was verified to ensure the black leg faced the cathode side.

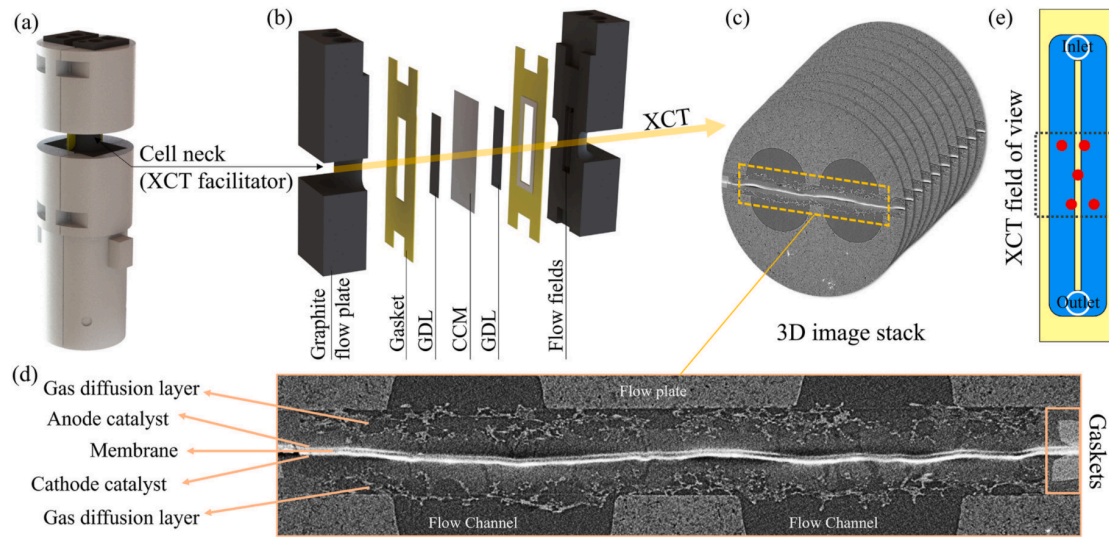


Figure 2: XCT-compatible PEM fuel cell assembly used in this project.



Figure 3: Fully assembled PEM fuel cell

2.3 Final Wiring and Pre-Mounting Check

Prior to mounting, all wire connections were visually and mechanically inspected:

- Ring terminals were crimped securely and fastened tightly.
- Thermocouple and heater terminals were routed to prevent tangling or stress.
- All three cathode (red) and anode (black) wires were checked for correct polarity and secure connection.

2.4 Mounting to Test Platform

The assembled fixture was mounted on the G20 station platform:

- Sensor cables and thermocouples were routed carefully to avoid mechanical interference.
- Foam insulation and Kapton tape were applied to exposed joints to minimize heat loss and electromagnetic interference.
- Gas fittings were installed in the following order:
 1. Supply and purge gas lines
 2. Outlet line (observed for bubbling in water)
 3. Return gas line (to complete closed-loop configuration)



Figure 4 : G20 Fuel Cell Test Station Setup at FCREL PowerTech Labs

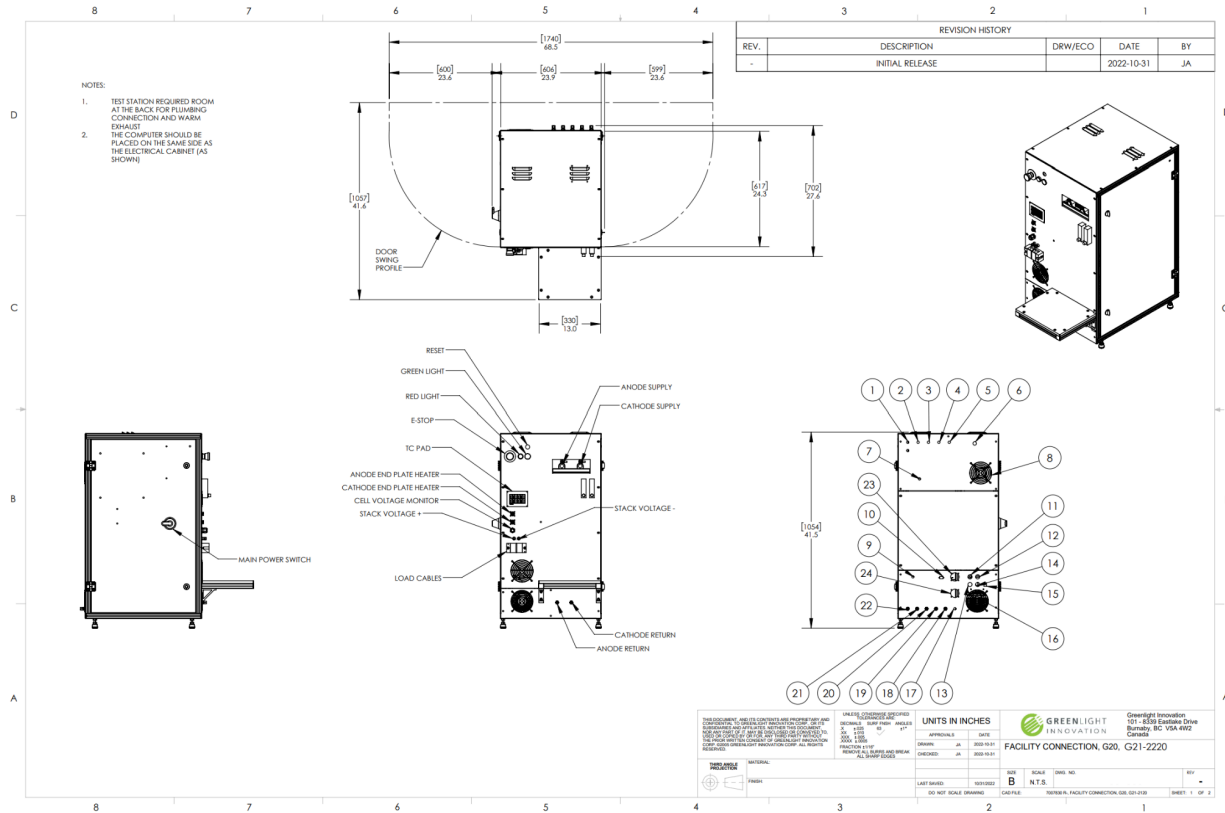


Figure 5: Facility Connection and Interface Layout of Greenlight G20 Station (G21-2220)

2.5 Electrical Setup and Software Configuration

The system was connected to the G20 control hardware and software:

- Thermocouples and heater wires were connected to designated ports.
- Anode and cathode ring terminals were connected to the potentiostat.
- The Emerald FCATS software was launched and set to the “Startup” workspace.
- Manual control was enabled via “Override Off” and adjustments were made in the “Pressure/Temperature” tab.
- Voltage polarity and channel mapping were cross-verified with the software configuration.

2.6 Leak Detection Procedure

Leak detection was performed on all fittings between the gas sources — including the external H₂ cylinder pack and the N₂ Dewar system — and the test fixture to verify integrity before hydrogen introduction.

- Leak detection fluid was applied to all gas fittings.
- 100% nitrogen was purged through both anode and cathode sides.
- Bubble formation was visually inspected.
- If no leaks were found, the system was switched to 100% hydrogen (anode side only).
- The Bacharach Leakator Jr. was used to scan for hydrogen leaks:
 - Device was calibrated in ambient air.
 - A slow, deliberate scan was done across all joints.
 - High-frequency beeping indicated hydrogen presence, prompting corrective action if needed.

2.7 FCATS Test Preparation

Once the system was leak-free, test setpoints were configured:

- Initial purge:
 - Anode and Cathode: 100% N₂
- Operating gases:
 - Anode: 100% H₂
 - Cathode: Ambient Air
- Flow, pressure, and stoichiometry setpoints were adjusted per cell size.

- Target temperatures:
 - Anode inlet: 40 °C
 - Cathode inlet: 40 °C
- Heaters were activated, and thermocouple feedback was monitored to verify temperature response.

2.8 Test Execution and Monitoring

Two main scripts were executed via the FCATS interface:

1. Startup Conditioning
2. MEA Conditioning

During the test, the following parameters were continuously monitored:

- Flow setpoint vs actual
- Pressure drop
- Inlet and outlet relative humidity (RH)
- Cell voltage and current (via potentiostat)
- Temperature distribution (via thermocouples)

Data was saved in .CSV format and stored for later analysis.

3. Results and Observations

This section presents the observed output and raw data collected during the test procedures performed on the single-cell PEM fuel cell. All data was acquired via the Greenlight G20 system and logged using the Emerald FCATS software. Test scripts executed included “Startup Conditioning” and “MEA Conditioning,” with output signals monitored through potentiostat channels.

3.1 General Assembly Outcome

The assembly was completed with all thermocouples, ring terminals, and heater wires properly connected and insulated. Leak detection confirmed that the system was sealed at all fittings and pipe joints. Upon initializing the Emerald-FCATS software, both temperature and gas flow controls responded as expected, and baseline readings were within operational limits prior to hydrogen flow introduction.

3.2 Voltage and Current Density Data

To illustrate the performance trends observed during testing, representative data from two random individual test cells — Cell 1 and Cell 2 — are presented below. These examples were selected for clarity and are used here to demonstrate the typical behavior of the PEM fuel cell during conditioning cycles.

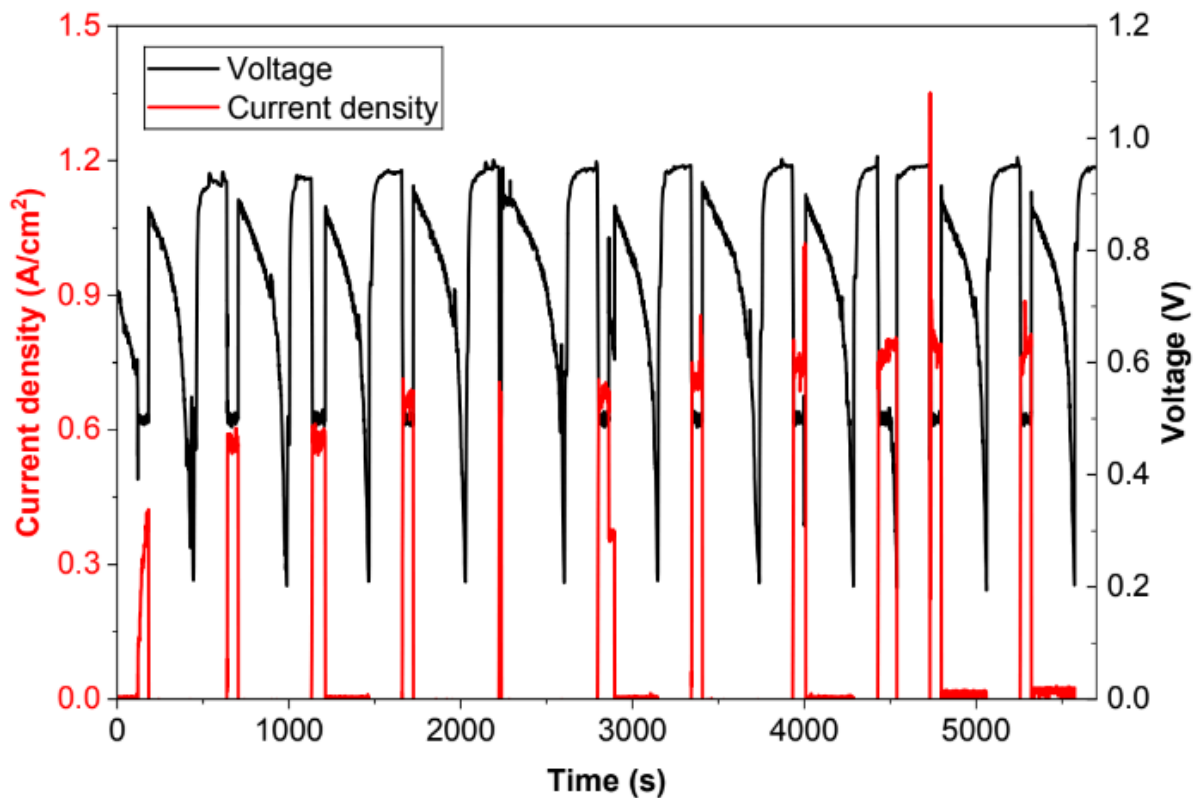


Figure 6: Representative voltage and current density data from Cell 1 used for illustration.

Cell 1: Voltage vs. Current Density

- The *graph on page 1* of the uploaded file shows voltage (black curve) and current density (red curve) plotted over time.
- Voltage ranges from approximately 0.6 V to 1.0 V during load cycles.
- Current density reaches a maximum of $\sim 0.6 \text{ A/cm}^2$ during peak periods.
- Repeated cycling demonstrates the cell's ability to stabilize across test intervals.

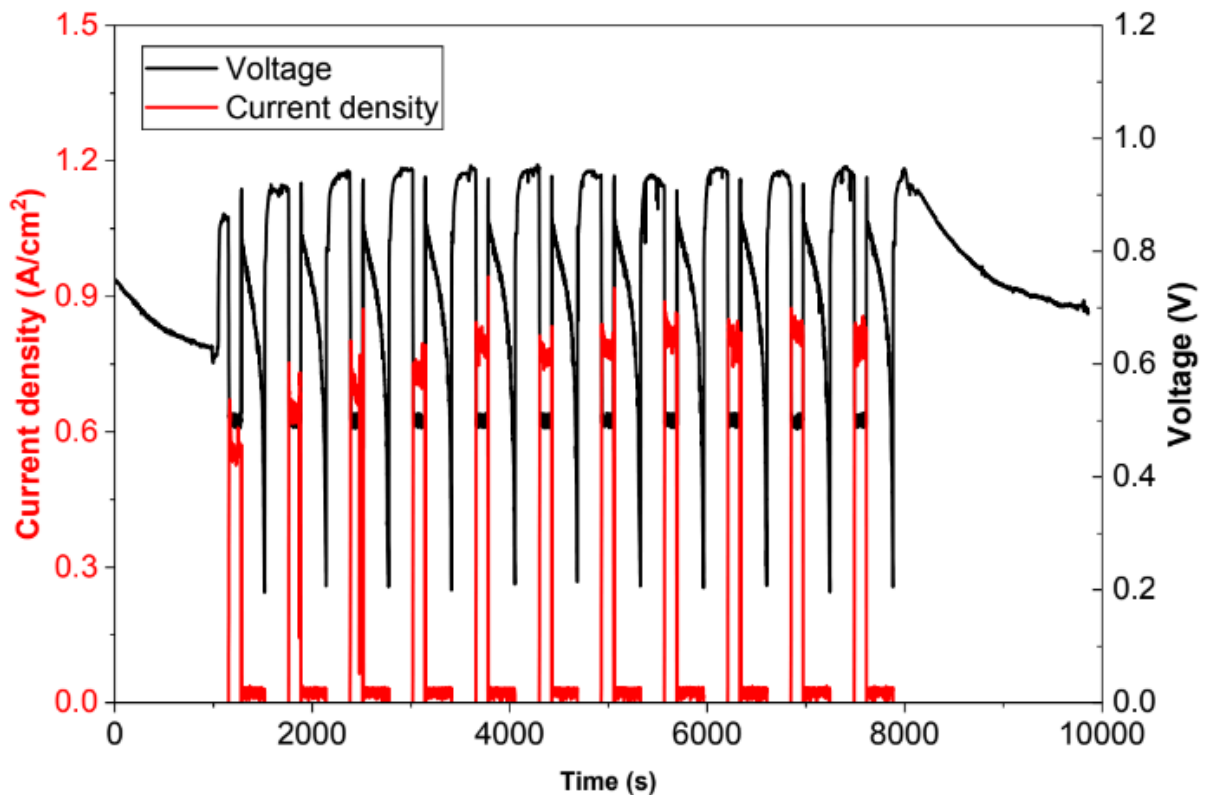


Figure 7: Representative voltage and current density data from Cell 2 used for illustration.

Cell 2: Voltage vs. Current Density

- As shown in the *graph on page 2*, Cell 2 also underwent multiple load cycles with consistent waveform patterns.

- Voltage behavior closely resembles Cell 1, with peak voltages around 0.9–1.0 V and a gradual decline after ~8000 seconds.
- Current density reaches up to ~0.6 A/cm² in each pulse, followed by rest phases with 0 A/cm².
- The data confirms the cell's electrochemical activity and responsiveness under controlled operating conditions.

3.3 Environmental and Operating Parameters

The table below summarizes the typical environmental setpoints and observed operating ranges recorded during the test session. These values correspond to the same representative data sets shown for Cells 1 and 2 in Section 3.2 and are provided to illustrate the expected system behavior under standard conditioning conditions. Full test data covering all cells is archived internally.

Parameter	Setpoint	Observed Range
Anode Temperature	40 °C	39.6–40.2 °C
Cathode Temperature	40 °C	39.5–40.1 °C
Anode Flow (H ₂)	100 sccm	99–101 sccm
Cathode Flow (Air)	200 sccm	198–202 sccm
Relative Humidity (inlet)	95%	93–96%

Pressure Drop (cell)	< 0.5 psi	0.3–0.45 psi
Cell Voltage (OCV)	—	1.02–1.04 V
Voltage during load	—	0.6–0.95 V

All recorded values fell within expected operational tolerances.

3.4 Data Logging and Storage

- All test data was exported as .CSV files and saved under the project directory on the lab's server.
- Voltage and current readings were sampled every second and timestamped for reproducibility.
- Figures above were generated from representative runs.

The figures shown in Section 3.2 were generated from two representative test cases; all test session logs are preserved for internal review and full-data analysis.

4. Discussion

This section interprets the observed behaviors and assesses their significance to the project objectives (procedure reliability, safety, and data quality)—separate from the Results per SFU's guideline that *Discussion explains/interprets results, not repeats them*.

4.1 Electrical Behavior

- **Stable operation:** OCV in the expected range and reproducible load cycles indicate correct polarity, sound contacts, and adequate gas supply control.
- **Gradual voltage decline (late run):** The slow drop after prolonged cycling is consistent with hydration imbalance or localized flooding/thermal drift. Given RH near setpoint and small ΔP , membrane hydration dynamics are the likely driver; targeted humidity sweeps

would confirm.

4.2 System Controls and Safety

- **Temperature/flow control:** Tight tracking to setpoints shows heaters, MFCs, and feedback loops are functioning as configured; warm-up lag was minimal once steady state was reached.
- **Leak integrity:** Zero bubble/leak-detector alarms through N₂ then H₂ checks validates the assembly sequence and joint selection for this fixture.

4.3 Data Quality & Reproducibility

- **Logging:** 1 Hz FCATS logging captured transients adequately for this scope. Channel mapping and polarity verification prevented sign errors; repeated pulse profiles across cells suggest the procedure is portable.

4.4 Limitations

- Single-cell, short-duration tests; no full polarization curves/EIS and limited environmental sweeps (temperature/RH/pressure). Sensor calibration was verified functionally but not traceable; future runs should include pre/post calibration checks.

4.5 Implications

- The standardized build-mount-verify workflow produced safe startup and repeatable data, meeting the immediate objective of creating a baseline procedure for subsequent MEA studies. Follow-on experiments should vary RH/temperature systematically and add diagnostics (polarization, EIS) to isolate hydration effects and quantify performance margins.

5. Conclusions & Recommendations

Conclusions

- A standardized build–mount–verify procedure was implemented for the single-cell PEM fixture on the G20.
- N₂ and H₂ leak checks passed; temperature/flow controls tracked setpoints during startup and conditioning.
- FCATS logging (1 Hz) captured all required channels; channel mapping/polarity were verified.
- Load-cycle runs produced stable, repeatable voltage–current behavior across cells within the tested window.
- The procedure establishes a safe, reproducible baseline for future MEA studies.

Recommendations

- Automate pre-run checks (sensor mapping, polarity, leak test) via a short FCATS “pre-flight” script and checklist.
- Expand the test matrix with RH/temperature/back-pressure sweeps; add polarization curves and EIS for diagnostics.
- Improve thermal insulation/cable strain relief to reduce warm-up lag and stabilize temperature gradients.
- Version-control FCATS scripts and standardize data export templates for repeatability.
- Record a simple safety sign-off (leak-test log + E-stop verification) per run.

6. References

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